Introduction to Machine Learning

Machine learning is a field of artificial intelligence that focuses on developing systems that can learn from and make decisions based on data. Instead of being explicitly programmed to perform a task, these systems improve their performance over time through experience.

Core Concepts in Machine Learning

Machine learning algorithms can be broadly categorized into several types:

Supervised Learning: Algorithms learn from labeled training data to make predictions or decisions. Examples include classification and regression problems.

Unsupervised Learning: Algorithms find patterns in unlabeled data. Common applications include clustering, dimensionality reduction, and anomaly detection.

Reinforcement Learning: Agents learn to make sequences of decisions by receiving feedback from their actions and environment.

Semi-supervised Learning: Uses both labeled and unlabeled data for training.

Popular Machine Learning Algorithms

- Linear and logistic regression
- Decision trees and random forests
- Support Vector Machines (SVM)
- Neural networks and deep learning
- K-means clustering
- Principal Component Analysis (PCA)

Applications of Machine Learning

- Computer vision and image recognition
- Natural language processing
- Recommendation systems
- Fraud detection
- Autonomous vehicles
- Medical diagnosis
- Financial forecasting